

Ninja SEO Analyzer — System Architecture & Product Status

Date: 2026-05-21 **Audience:** Product & Engineering

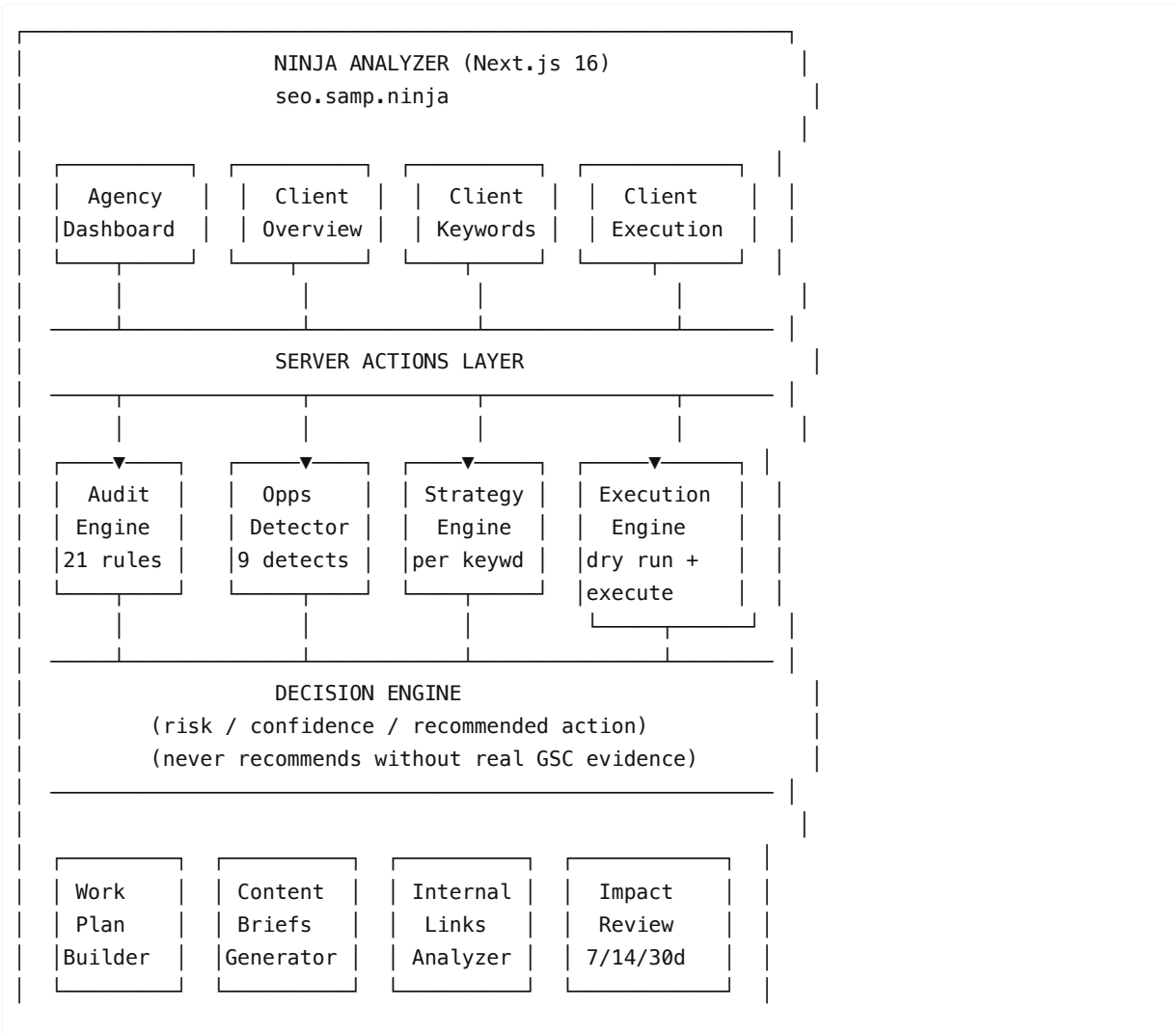
What Is This System?

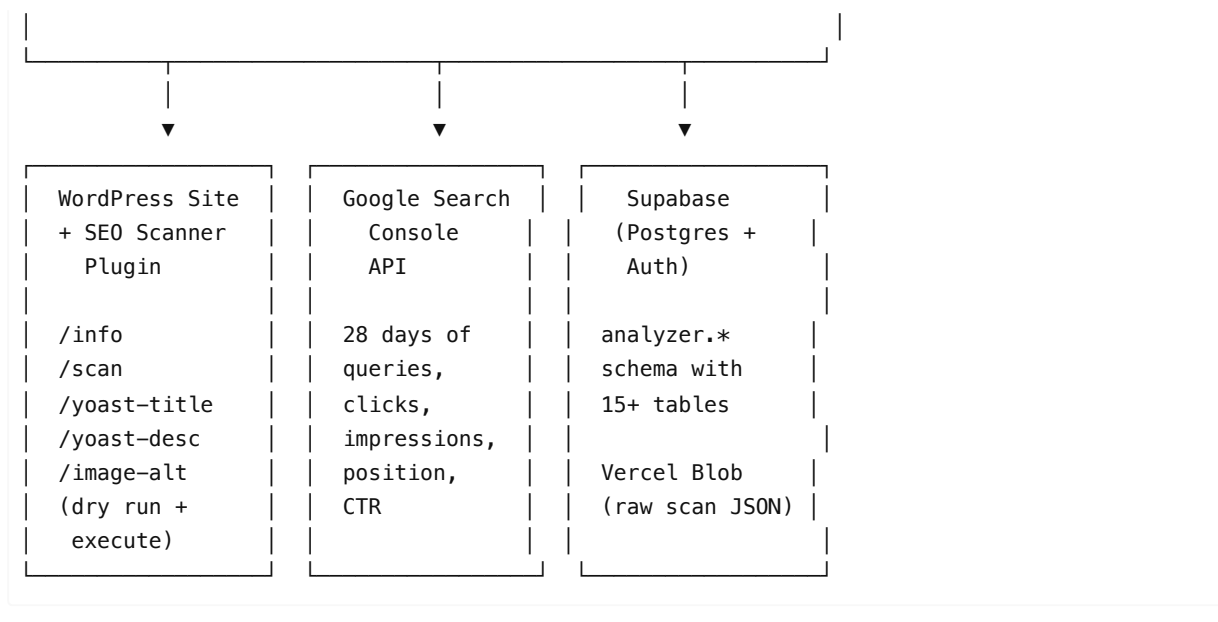
An internal tool for an SEO agency. The agency manages WordPress client websites. This system:

- 1. Connects to each client's WordPress site via a plugin
- 2. Analyzes the site's SEO health
- 3. Finds optimization opportunities using real Google Search Console data
- 4. Builds strategies per keyword
- 5. Generates work plans with grouped, risk-scored actions
- 6. Executes changes on the WordPress site (with human approval + dry run + rollback)
- 7. Measures impact over time (7/14/30 days)

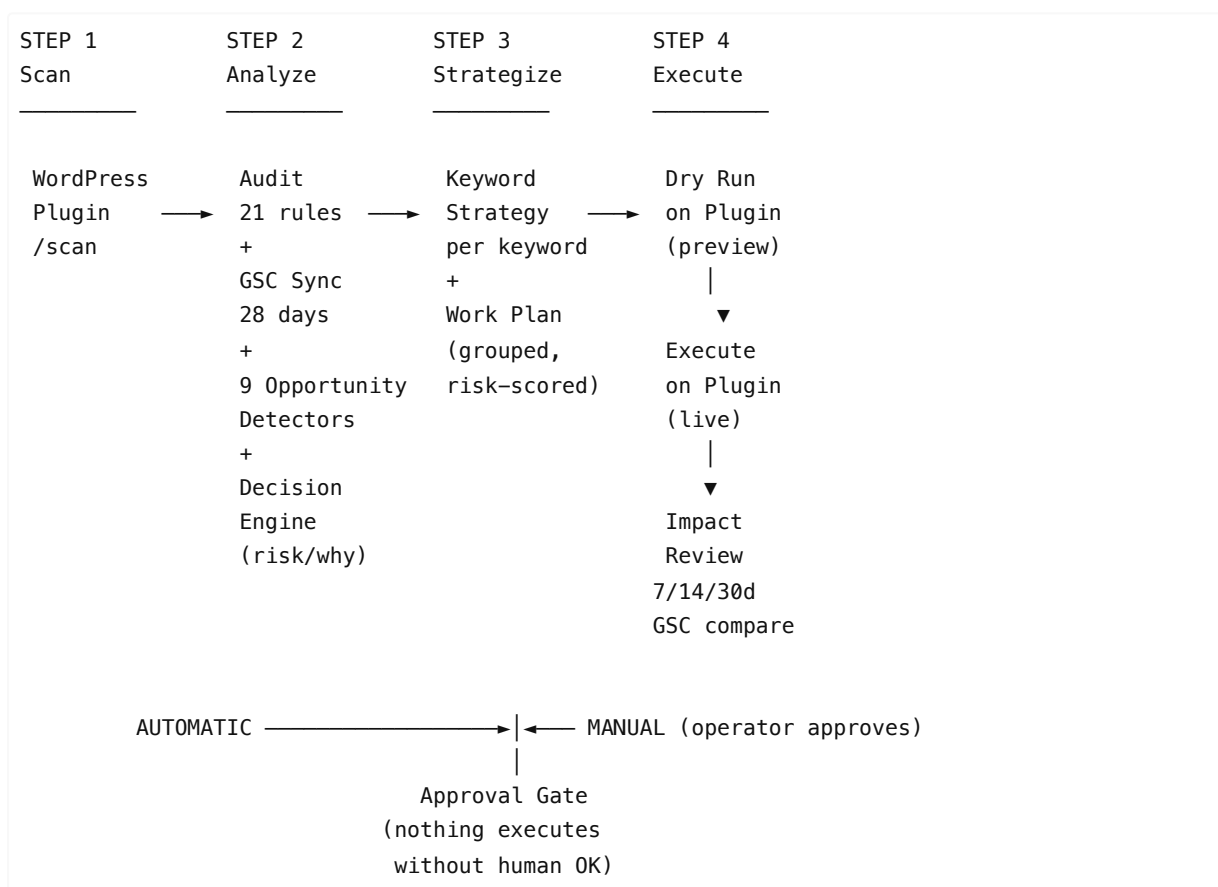
Live at: `seo.samp.ninja`

Architecture Overview





The Pipeline — Step by Step



Step 1 — Scan (connects to the WordPress site)

The client has a WordPress plugin (`agency-seo-scanner`) installed on their site. When the operator triggers a scan:

- The system calls the plugin's `/scan` endpoint
- The plugin returns every page on the site with full SEO data:

- Yoast SEO title, meta description, focus keyword
 - Heading structure (H1, H2, H3...)
 - Word count, images, internal/external links
 - Featured image, alt texts
 - WooCommerce product data (if applicable)
 - Canonical URLs, robots directives
 - Schema markup
- This raw JSON is stored in Vercel Blob (cloud storage)
- The scan metadata (size, duration, summary) is stored in Postgres

Step 2 — Analyze (find what's wrong + what's possible)

Three things happen after a scan (or on a scheduled refresh):

A. Audit Engine — 21 rules across 8 categories check the scan data:

Category	What it checks
Indexing	noindex on content, external canonicals, canonical mismatches
On-Page Meta	Missing title/description/focus keyword, length violations
HTML Structure	Missing H1, multiple H1s, heading hierarchy skips
Content Quality	Thin content (<300 words)
Images	Missing alt text, generic filename-as-alt, featured image alt
Internal Links	Orphan pages (no incoming links), pages with no outgoing links
Schema	Missing schema type, products without schema
Cannibalization	Duplicate focus keywords, duplicate titles across pages

Each rule produces a **Finding** with severity (critical / important / minor / info), affected page count, and a fix hint.

B. Google Search Console Sync — Pulls the last 28 days of real search data:

- Which queries bring traffic to which pages
- Clicks, impressions, CTR, average position per query
- This is the ground truth — what Google actually shows users

C. Opportunity Detection — 9 detectors cross-reference GSC data with audit findings:

Detector	What it finds
Low CTR	Good position but people don't click (title/description problem)
Quick Win Position	Position 6-15, high volume — small push could reach top 3
High Impressions No Clicks	Google shows the page but nobody clicks (snippet issue)
Declining Clicks	Traffic dropping (competitor overtaking?)
Declining Position	Position falling (quality/relevance issue)
Target Keyword Not Ranking	Keyword in bank but site doesn't appear in Google for it
Target Keyword Needs Optimization	Ranks but underperforms (wrong page, weak content)

Cannibalization	Multiple pages competing for the same query (diluting each other)
New Query Growth	New search demand appearing (opportunity to capture early)

Each opportunity gets a **Decision** computed by the Decision Engine:

- **Risk level:** low / medium / high / critical
- **Confidence:** based on data quality and quantity
- **Recommended action:** execute / review / monitor / research / skip
- **Why:** A concrete, data-grounded explanation (e.g., "Position 8, 1,200 monthly searches, CTR 1.2% vs expected 5% — title update could 3x clicks")

Key rule: The engine will **never** recommend execution without a substantive, GSC-grounded reason. If it can't justify the action with real numbers, it flags for human review.

Step 3 — Strategize (plan the work)

A. Keyword Strategy — For each target keyword in the bank:

- Builds a performance snapshot (current position, clicks, CTR, trend)
- Classifies into a strategy type:
 - `protect_position` — Already top 3, don't touch, monitor
 - `quick_win` — Position 6-10 with CTR gap, title/meta refresh
 - `content_boost` — Needs deeper content
 - `internal_link_boost` — Needs more internal links pointing to it
 - `new_content_needed` — No page ranks for this keyword, create one
 - `cannibalization_fix` — Multiple pages fighting, consolidate
 - `monitor_only` — Not enough data yet
- Generates a 3-5 step action plan per keyword
- Includes measurement plan (what to check at 7/14/30 days)

B. Content Briefs — For opportunities/strategies that need content work:

- Generates structured briefs: recommended title, meta description, H1, content outline, secondary keywords, internal links, CTA, schema
- Intent-aware: transactional keywords get product-focused briefs, informational get guide-style outlines
- Hebrew-aware title construction

C. Internal Link Suggestions — Analyzes the site's link graph:

- Finds orphan pages that need incoming links
- Finds high-authority pages that should link to target pages
- Suggests anchor text from keyword bank
- Prioritizes by impact and confidence

D. Work Plan — Groups everything into a single actionable plan:

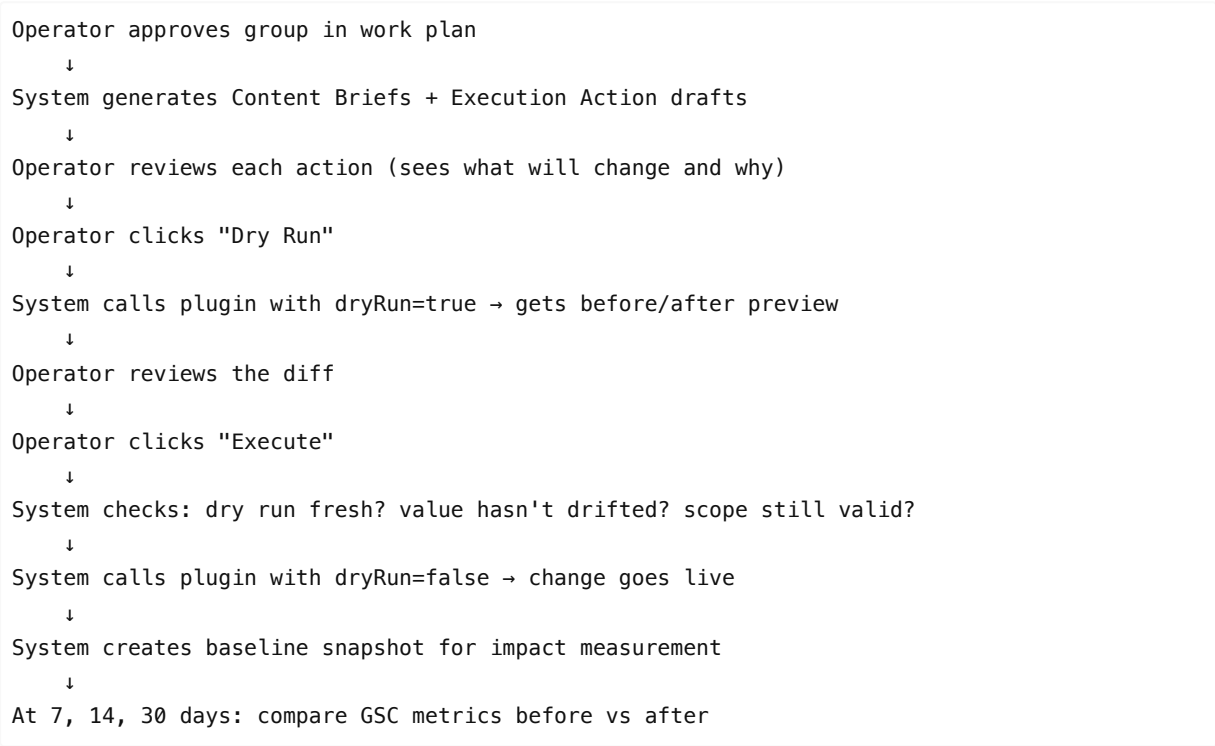
Group	What's in it	Risk
safe_meta	Low-risk title/description updates on non-critical pages	Low
quick_wins	Position improvements with clear evidence	Low-Medium
content_expansion	Content additions/rewrites on existing pages	Medium
internal_linking	Link insertions with high confidence	Low
human_review	Anything the engine isn't confident about	Varies

blocked	Can't execute (scope, technical issues)	—
monitor_only	Watch and wait, not enough signal	—

Each group has a cap (e.g., max 30 safe_meta items) to avoid overwhelming the operator.

Step 4 — Execute (make changes on the site)

This is the only part where changes are made on the actual WordPress site. It's heavily gated:



What can be executed today:

- Yoast title updates
- Yoast meta description updates
- Image alt text updates

Preview-only (no auto-execute):

- Internal link insertions
- Content snippet insertions

Safety features:

- Nothing executes without dry run first
- Dry runs expire after 24 hours (must re-run if stale)
- Drift detection (if someone manually edited the field since dry run, execution is blocked)
- Rollback available for title/description/alt changes
- Rollback checks for drift too (won't revert if value was manually changed)

Product Requirements vs Current State

1. "Go over all meta on the site — what's there, what's missing per page"

STATUS: FULLY BUILT

The Audit Engine runs 21 rules on every scan. Per-page breakdown shows exactly which pages are missing titles, descriptions, focus keywords, H1s, schema, etc. Results are grouped by category with severity ratings.

2. "Internal links — where to add, where not"

STATUS: FULLY BUILT

The Internal Link Analyzer identifies orphan pages, pages without outgoing links, and suggests specific source→target link pairs with recommended anchor text. Suggestions are prioritized by impact and filtered through the scope engine (won't suggest links on cart/checkout/legal pages).

3. "Write blog posts / change content / add to product pages"

STATUS: PARTIALLY BUILT

What exists:

- Content Brief generator — produces structured briefs with title, outline, keywords, CTA, schema recommendations
- Brief types: new article, new landing page, optimize existing page, expand content, FAQ section, title/meta update

What's missing:

- **Actual content generation (AI writing)** — the system produces the brief/outline but doesn't write the actual text. An AI writer (Claude API or similar) would take the brief and produce publishable content.

4. "Google Search Console for keyword progress tracking"

STATUS: FULLY BUILT

Central Google OAuth account syncs the last 28 days of search data per client. Tracks queries, pages, clicks, impressions, CTR, position. Used by all engines (opportunities, decisions, strategies, impact reviews) as ground truth.

5. "Keyword bank — client gives me keywords, system works, I just approve the plan"

STATUS: FULLY BUILT

The TargetKeyword model stores the keyword bank per client. When keywords are added:

1. System computes a strategy per keyword (protect / quick win / content boost / etc.)
2. Builds an action plan (3-5 steps)
3. Groups everything into a work plan
4. Operator approves groups → system prepares briefs and execution actions
5. Operator reviews and executes

This is exactly the flow described: keywords in → system works → operator approves.

6. "Blog content calendar — timed, researched, not 100 posts at once"

STATUS: NOT BUILT

What's missing:

- **ContentCalendar model** — scheduling when each piece of content goes live
- **Publishing cadence** — configurable rate (e.g., 2-3 posts/week)
- **Content dependencies** — cornerstone content before supporting posts
- **Seasonal/trend awareness** — timing content to search trends
- **Queue management** — draft → review → scheduled → published pipeline

7. "Quality score for site speed / code improvement"

STATUS: NOT BUILT

What's missing:

- **PageSpeed Insights integration** — Google's free API, returns Core Web Vitals (LCP, FID, CLS), performance score, specific recommendations
- **Per-page speed scoring** — track over time, show improvement/regression
- **Code quality recommendations** — image optimization, lazy loading, render-blocking resources, etc.

The Missing Piece: How the Operator Actually Interacts

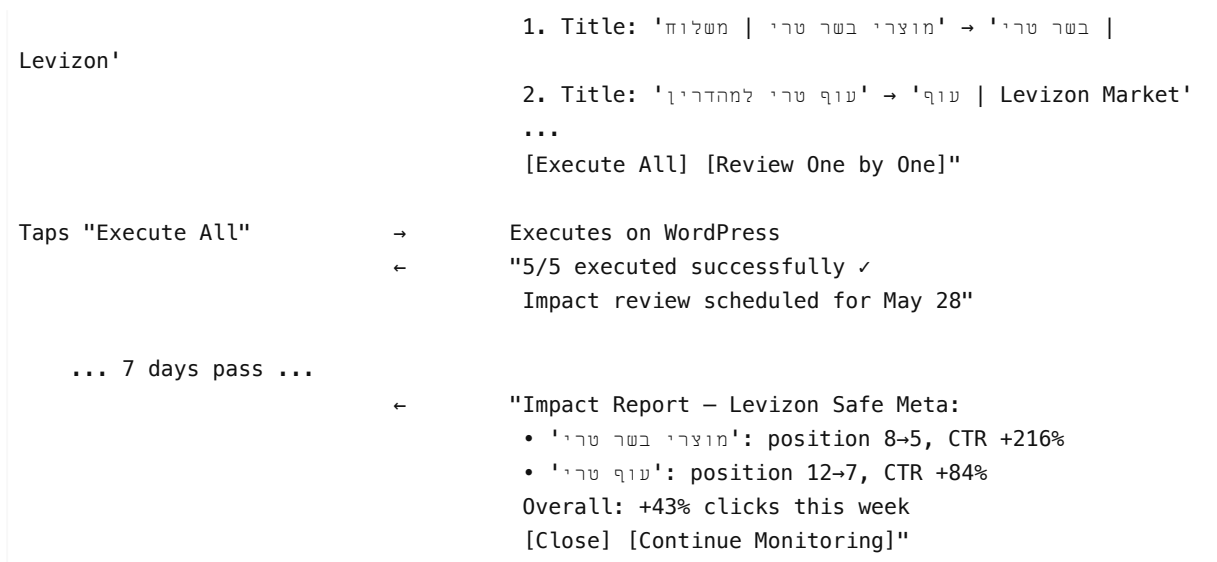
Current model: Dashboard (operator drives)

Operator		System
Opens browser		
Navigates to client	→	Shows data
Clicks "Run Scan"	→	Scans site
Goes to Opportunities tab	→	Shows list
Reviews one by one	→	Waits
Clicks "Approve"	→	Prepares
Goes to Execution tab	→	Shows drafts
Clicks "Dry Run"	→	Runs preview
Reviews diff	→	Waits
Clicks "Execute"	→	Makes change
Comes back in a week	→	Has impact data

Problem: The operator has to remember to check, navigate 14 tabs per client, and manually trigger every step. The engines are powerful but the interaction model is "pull" — you only see results when you go looking.

Target model: Notification + Approval (system drives)

Operator		System
Adds keywords for client	→	Receives keywords ↓ automatically: Syncs GSC Runs audit Detects opportunities Computes strategies Builds work plan
	←	"Here's the plan for Levizon: 5 safe meta updates 3 quick wins 2 content pieces needed [Approve Safe Meta] [Approve Quick Wins] [Review All]"
Taps "Approve Safe Meta"	→	Generates briefs Creates execution drafts Runs dry runs automatically
	←	"5 dry runs ready:"



The operator's job becomes: **add keywords, approve plans, review results.** Everything else is automated.

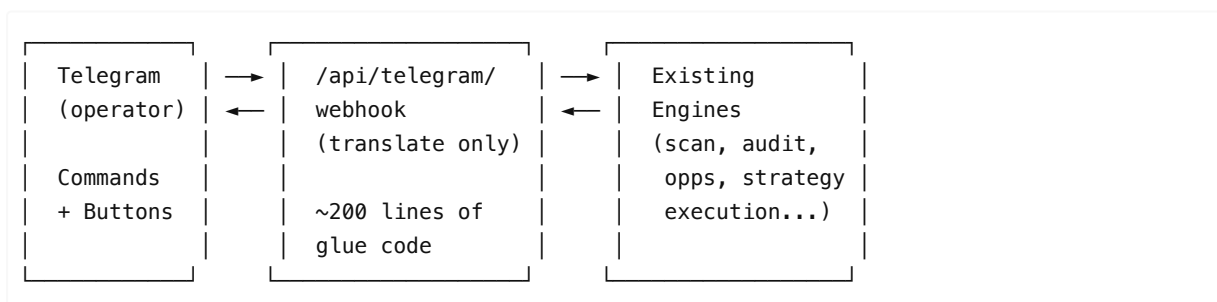
How to deliver this: Telegram Bot

A Telegram bot is the simplest way to deliver the notification + approval model:

- **Free**, no business verification, no message limits
- **Inline keyboards** — unlimited buttons for approve/reject/review
- **Proactive messages** — system can notify anytime (no 24-hour window like WhatsApp)
- **Mobile-first** — operator works from their phone
- **Simple API** — one webhook endpoint, one send function

The bot is a **thin translation layer** — it doesn't contain business logic. It just:

1. Receives operator commands → calls existing server functions
2. Receives engine results → formats as Telegram messages with buttons



The daily operator experience (target)

Morning:

- Bot sends daily digest: "3 clients need attention, 2 impact reviews ready"
- Operator taps a client → sees summary → approves or skips

When keywords are added:

- Operator sends keywords in chat (or adds via web UI)
- Bot responds within minutes with a strategy per keyword
- Operator approves the plan → system works

When things happen:

- Scan finishes → bot notifies with summary
- Opportunities detected → bot shows top ones with approve buttons
- Execution completes → bot confirms
- Impact review ready → bot shows before/after metrics

The web dashboard still exists for deep-dives, reports, and detailed review. But the daily workflow moves to Telegram.

Summary: What to Build Next

Priority	Feature	What it unlocks	Effort
1	Telegram Bot — notification + approval flow	Operator works from phone, system drives	3-4 days
2	Auto-pipeline on keyword add — adding a keyword triggers GSC → strategy → brief → plan automatically	"I add keywords, system works"	1-2 days
3	AI Content Writer — take existing briefs, produce actual content via Claude/GPT API	Blog posts and content pages	3-4 days
4	Content Calendar — schedule publishing with cadence control	Timed content, not 100 posts at once	2-3 days
5	PageSpeed Integration — Google PSI API, per-page scores, tracking	Speed quality score	1-2 days

Total: ~12-15 days to fully deliver all 7 product requirements + the interaction model.

Items 1-2 change **how** the operator works (from dashboard-babysitting to approve-from-phone). Items 3-5 add **what** the system can do (write content, schedule it, measure speed).

Tech Stack

Component	Technology
App framework	Next.js 16 (App Router)
Frontend	React 19, Tailwind 4, Hebrew RTL
Database	PostgreSQL on Supabase (EU, Frankfurt)
ORM	Prisma 7 with pg adapter
Auth	Supabase Auth (email/password, admin_users gate)
Storage	Vercel Blob (scan payloads)
Hosting	Vercel (Frankfurt region)
WordPress connector	Custom plugin agency-seo-scanner
External APIs	Google Search Console, Google OAuth
Language	TypeScript throughout

Data Model (simplified)

Client

```
|— Scan[] ————— Finding[]
|— TargetKeyword[] — KeywordStrategy[]
|— Opportunity[] ——— ExecutionAction[] ——— ExecutionEvent[]
|— ContentBrief[]
|— InternalLinkSuggestion[]
|— GscDailyRow[]
|— ImpactBaseline[] — ImpactReview[]
|— SeoWorkPlan[] ——— SeoWorkPlanItem[]
|— AutomationRun[]
```

GscAccount (singleton – one Google account for the whole agency)